

AI Workflow Audit & Tool Setup (FL-01)

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1. Weekly Workflow Audit (12 Genuine Tasks)

Recurring tasks across systems engineering (Zephyr RTOS), machine learning research (FlyRank), full-stack web (Next.js), open source, and academic studies classified using Ethan Mollick’s AI delegation framework.

#	Task Name & Context	Classification	One-Line Strategic Rationale
1	Hardware ISR Design Embedded RTOS (Zephyr)	Just Me	Real-time race conditions, register timings, and hardware deadlocks require human spatial reasoning; hallucinations brick boards.
2	Research Hypothesis Framing Machine Learning (FlyRank)	Collaborate	Iterative rubber-ducking accelerates trade-off analysis between capacity limits, metrics, and non-linear decay patterns.
3	TypeScript API Schemas Next.js / GeetHub Engine	Delegate w/ Review	Generating type contracts from sample JSON is fast and mechanical, needing only a brief syntax and nullability check.
4	Career Retrospectives Professional Growth	Just Me	Outsourcing personal introspection, self-critique, and goal prioritization destroys authentic cognitive reflection.
5	Data Pipelines (DuckDB/Pandas) Search Analytics / FlyRank	Delegate w/ Review	Complex multi-index aggregations are drafted rapidly by AI, but must be verified against actual dataset grains.
6	GSAP Animation Choreography Frontend UI (Zafran / GeetHub)	Collaborate	Conversational iteration allows rapid prototyping of bezier ease curves, scroll triggers, and DOM state choreographies.
7	Codebase Linting & AST Checks Dev Environment / Git Hooks	Fully Automate	Pre-commit hooks (ruff, prettier) execute deterministically in milliseconds without cognitive load or AI token cost.
8	Concurrency Bug Investigation Distributed Systems / Kernels	Collaborate	Providing error traces and lock sequences allows systematic enumeration of edge cases and lock-ordering hazards.
9	PR Changelogs & Summaries Open Source (Apicurio Registry)	Delegate w/ Review	Condensing unified git diffs into structured markdown changelogs saves hours, requiring only technical sanity checks.
10	Core DSA / Algorithm Problems Academic CS Coursework	Just Me	Relying on AI solutions prevents building the deep internal neural pathways required for first-principles problem solving.
11	Adversarial Mock Test Cases Software QA & Security	Collaborate	Prompting AI for adversarial edge-case payloads reveals subtle structural blind spots human engineers routinely overlook.
12	CI Smoke Tests & Data Guards GitHub Actions Workflows	Fully Automate	Rule-governed CI workflows enforce test suites and dataset leak checks deterministically on every push.

2. Three Target Tasks for FL-02 through FL-04

Target Task 1: Exploratory Feature Engineering & Hypothesis Formulation (Collaborative)

- **Domain:** Machine Learning Research & Tabular Analytics (FlyRank Track)
- **What 'Done Well' Means:** (1) Produces ≥ 4 non-trivial candidate features from column interactions; (2) 100% free of target leakage (excludes any field derived from target outcome window); (3) Explicit operational justification tying each signal to human/search behavior; (4) Measurable correlation gain or tree feature importance without degrading holdout Precision@50.

Target Task 2: Type-Safe API Data Contracts & Validation Interfaces (Delegate w/ Review)

- **Domain:** Full-Stack Architecture & Distributed Systems (Next.js / Apicurio)
- **What 'Done Well' Means:** (1) Zero usage of any; strict optionality and union discrimination; (2) Emits accompanying runtime parsing schemas (Zod) validating valid/invalid edge-case fixtures; (3) Reduces boilerplate drafting time from 30 min to <2 min with ≤ 1 human adjustment during code review.

Target Task 3: Concurrency Vulnerability Analysis & Architectural Rubber-Ducking (Collaborative)

- **Domain:** Embedded Firmware & Distributed Systems (Zephyr RTOS)
- **What 'Done Well' Means:** (1) Systematically enumerates ≥ 2 non-obvious deadlock or data-race scenarios; (2) Uses authentic RTOS synchronization primitives (mutexes, spinlocks, work queues) respecting ISR constraints; (3) Proposes concrete, minimal code diffs that resolve race conditions without latency jitter.

3. Claude Project Configuration & Academy Evidence

Project Name: Engineering & ML Research Co-Pilot
Tone & Profile: Treat as senior technical peer. Concise, direct, first-principles, zero boilerplate greetings. Focus on trade-offs (latency vs.

memory, in-sample fit vs. holdout generalization). Adhere strictly to data safety (no PII, no unvetted leakage) and frame ML outputs as directional decision-support.

Academy Enrollment: Enrolled in *AI Fluency: Framework & Foundations* (Anthropic Skilljar). Completed Module 1.